



Research Interests

I develop machine-learning co-designed quantum systems to discover novel architectures beyond human intuition, enabling more capable, scalable, and experimentally realizable quantum photonic technologies.

Professional Experience

2025 - current	Assistant Professor	University of Hawai'i, Mānoa (UHM), Electrical and Computer Engineering (ECE), Honolulu, HI
2023 - 2025	Postdoctoral Researcher	Massachusetts Institute of Technology (MIT), Research Laboratory of Electronics (RLE), Cambridge, MA Advisor: Prof. Dirk Englund
2022	Visiting Student	University of Michigan, Electrical Engineering and Computer Science (EECS), Ann Arbor, MI Advisor: Prof. Zheshen Zhang
2014 - 2016	Research Assistant	Academia Sinica, Institute of Atomic and Molecular Sciences (IAMS), Taipei, Taiwan Advisor: Dr. Ying-Cheng Chen

Education

2017 - 2022	Ph.D. , <i>Physics</i> ,	The University of Arizona (UArizona), Tucson, AZ, USA Advisor: Prof. Zheshen Zhang Thesis: Continuous-variable photonic quantum information processing [Link]
2011 - 2014	M.S. , <i>Physics</i> ,	National Tsing Hua University (NTHU), Hsinchu, Taiwan Advisor: Prof. Ite Yu Thesis: Studies of Transient Effects from Single- Λ to Double- Λ Systems under the All-Resonant Condition [Link]
2007 - 2011	B.S. , <i>Electrophysics</i> ,	National Chiao Tung University (NCTU), Hsinchu, Taiwan

Grants Experience

2026	B.-H. Wu (Lead PI, UHM) and H. Choi (Co-PI, SBU), "Quantum Intelligent Sensor Network: Entanglement-Assisted Sensing and Machine-Learning Co-Design," NSF FET (25-543), \$600,000, Awarded [2026–2029] [Link] .
2026	H. Choi (Lead PI, SBU) and B.-H. Wu (Co-PI, UHM), "Loss-Tolerant Entanglement Protocols for Long-Distance Quantum Networks," NSF NeTS (25-543), \$750,000, Pending [2027–2030].
2026	Y. Zhang (PI), X. Xue (Co-PI), H. Tang (Co-PI), A. Kuh (Co-PI), D. Pavlovic (Co-PI), I. Molybog (Co-PI), L. Zhu (Co-PI), and B.-H. Wu (Co-PI), "Workforce for Resilient AI Platforms from Hawaii," NSF CyberAI SFS (26-503), \$2,299,698, Pending [2026–2029].

- 2024 D. Englund (Co-written by **B.-H. Wu**), “Silicon-Based Quasi-Deterministic Entangled Photon Pair Source for Zero-Added-Loss Multiplexing,” AFRL (FA8750-25-2-0500), \$700,000, **Awarded** [2024–2027].
- 2023 M. Yu (Lead PI, USC), Z. Chen (Co-PI, USC), D. Englund (Co-PI, MIT), and Q. Zhuang (Co-PI, USC) (Co-written by **B.-H. Wu**, MIT), “Squeezed Light Detector using Integrated Quadratically Nonlinear Photonic Molecules,” DARPA INSPIRED (HR001123S0052), \$3,943,979, **Awarded** [2024–2027].

Teaching Experience

- 2026 ECE470/PHYS460 (Physical Optics): UHM, Electrical and Computer Engineering and Physics and Astronomy – **Lecturer** [Link].
- 2025, 2026 ECE491D/693D (Introduction to Quantum Information Science): UHM, Electrical and Computer Engineering – **Lecturer** [Link].
- 2023 6.2410 (Quantum Engineering System): MIT, Electrical Engineering and Computer Science - Guest Lecturer.
- 2019 MSE350 (Numerical Methods): UArizona, Materials Science and Engineering - Guest Lecturer.
- 2018 PHYS241 (Introduction to Electricity and Magnetism): UArizona, Physics - Teaching Assistant.
- 2017-2018 PHYS141 (Introductory Mechanics): UArizona, Physics - Teaching Assistant.
- 2017 PHYS181 (Introductory Laboratory I): UArizona, Physics - Teaching Assistant.
- 2012 Mathematical and Scientific Thinking Lecture (general education): NTHU, Physics - Teaching Assistant.
- 2012 Daily Physics (general education): NTHU, Physics - Teaching Assistant.
- 2011 Introduction to Electrodynamics (I, II): NTHU, Physics - Teaching Assistant.

Patent

1. “Microring Perceptron Sensor for High-Speed, Low-SNR Radio-Frequency Signals,” **B.-H. Wu**, S.-Y. Ma , S. K. Vadlamani, H. Choi, D. Englund, U.S. Patent Application No.: 19/650,899.
2. “Quantum-computing architecture based on multi-dimensional continuous-variable cluster states in a scalable platform”, R. Alexander, S. Liu, **B.-H. Wu**, and Z. Zhang, U.S. Patent No. US12254381B1 [Link].

Awards and Honors

- 2026 OVPRS Travel Award: Funding support for presenting research at CLEO.
- 2012 Shang-Keng Ma Scholarship: Outstanding Statistical Mechanics study (Top 5%)
- 2010 Movie Physics Competition: Creating videos that analyzed the physics in movies (Top 5%)
- 2007, 2008 Joint Calculus Examinations (I, II): Calculus competition (Top 2%)

Professional Activities

1. **Workshop Panelist:** Scalable Information Processing with Quantum Nano-Photonics [Link] (2026).
2. **Session Chair:** Conference on Lasers and Electro-Optics (CLEO), Novel Quantum Processing (2026).
3. **Poster Judge:** UHM ECE 496 poster session competition (2025, 2026).
4. **Member:** American Physical Society (APS).
5. **Journal Reviewer:** Phys. Rev. Lett., Communication Physics, Phys. Rev. Research, Quantum Electronics, App Entropy, Optical and Quantum Electronics, IEEE JSAC, Phys. Rev. A., Appl. Opt.

Journal Publications

- 2026 1. “Bidirectional Nonlinear Optical Tomography: Unbiased Characterization of Off- and On-Chip Coupling Efficiencies,” **B.-H. Wu**, M. J. Mehrabad, D. Englund, Phys. Rev. Applied, 26, (1), 014053 [\[Link\]](#).
- 2026 2. “18-dB on-chip vacuum squeezing in an adaptively poled lithium niobate waveguide,” T. S. Karnik, X. Ren, C.-H. Lee, **B.-H. Wu**, *et al*, arXiv:2605.27607 [\[Link\]](#).
- 2026 3. “Quantum squeezing in an all-resonant periodically poled lithium niobate microresonator,” X. Ren, R. Kopparapu, T. S. Karnik, C.-H. Lee, K. Kwon, C. Cheung, Y. Yu, S.-Y. Ma, **B.-H. Wu** *et al*, arXiv:2602.22693 [\[Link\]](#).
- 2025 4. “Wafer-Scale Squeezed-Light Chips,” S. Liu, K. Zhou, Y. Zhang, A. Hariri, N. Reynolds, **B.-H. Wu**, Z. Zhang, arXiv: 2509.10445 [\[Link\]](#).
- 2025 5. “All-optical reconfiguration of single silicon-vacancy centers in diamond for non-volatile memories,” Y. Xue, X. Ni, M. Titze, S. S. Su, **B.-H. Wu**, L. Zhang, C. Cui, S. Guha, M. Eichenfield, L. Fan, Nat. Commun. 16, 6275 [\[Link\]](#).
- 2025 6. “Micro-Ring Perceptron Sensor for High-Speed, Low-Power Radio-Frequency Signal,” **B.-H. Wu**, S.-Y. Ma, M. J. Mehrabad, S. K. Vadlamani, H. Choi, D. Englund, arXiv:2504.16119 [\[Link\]](#).
- 2024 7. “Formation of Quasi-Bound States in the Continuum in a Single Deformed Microcavity,” S. Liu, **B.-H. Wu**, J. Huang, Z. Zhang, Photon. Res. 12, 2651 [\[Link\]](#).
- 2024 8. “Distinguishing under-and over-coupled resonances without prior knowledge,” C. Cui, L. Zhang, **B.-H. Wu**, S. Liu, P.-K. Chen, L. Fan, Optica 11 (2), 176 [\[Link\]](#).
- 2023 9. “Entanglement-assisted multi-aperture pulse-compression radar for angle resolving detection,” **B.-H. Wu**, S. Guha, Q. Zhuang, Quantum Sci. and Technol. 8 (3), 035016 [\[Link\]](#).
- 2022 10. “Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications,” **B.-H. Wu**, Z. Zhang, Q. Zhuang, Quantum Sci. and Technol. 7 (2), 025018 [\[Link\]](#).
- 2020 11. “Quantum computing with multidimensional continuous-variable cluster states in a scalable photonic platform,” **B.-H. Wu**, R. N. Alexander, S. Liu, Z. Zhang, Phys. Rev. Research 2, 023138 [\[Link\]](#).
- 2020 12. “Broadband coherent optical memory based on electromagnetically induced transparency,” Y.-C. Wei, **B.-H. Wu**, Y.-F. Hsiao, P.-J. Tsai, Y.-C. Chen, Phys. Rev. A 102 (6), 063720 [\[Link\]](#).
- 2019 13. “Theoretical study of a memory-based optical converter with degenerate Zeeman states,” P.-J. Tsai, Y.-C. Wei, **B.-H. Wu**, S.-X. Lin, Y.-C. Chen, Phys. Rev. A 100 (6), 063843 [\[Link\]](#).
- 2017 14. “Enhanced spectral profile in the study of Doppler-broadened Rydberg ensembles,” **B.-H. Wu**, Y.-W. Chuang, Y.-H. Chen, J.-C. Yu, M.-S. Chang, I. A. Yu, Sci. Rep. 7 (1), 9726 [\[Link\]](#).
- 2016 15. “High conversion efficiency in resonant four-wave mixing processes,” C.-Y. Lee, **B.-H. Wu**, G. Wang, Y.-F. Chen, Y.-C. Chen, I. A. Yu, Opt. Express 24 (2), 1008 [\[Link\]](#).

Conference Publications

- 2026 1. “Bidirectional Nonlinear Optical Tomography: An Unbiased Estimator for On-chip Squeezing,” **B.-H. Wu**, M. J. Mehrabad, D. Englund, CLEO, Charlotte, NC, USA.
- 2026 2. “Optimizing the Hardware as an AI Model: Physics-Aware Micro-Ring Sensor for High-Speed, Low-Power Signals,” S.-Y. Ma, **B.-H. Wu**, M. J. Mehrabad, Y. Han, S. K. Vadlamani, H. Wang, H. Choi, D. R. Englund, CLEO, Charlotte, NC, USA.

- 2026 3. “Vacuum Squeezing in Adaptively Poled Lithium Niobate Nanowaveguides,” T. S. Karnik, X. Ren, C.-H. Lee, M. Chaudhari, I. Christen, K. Kwon, K. Kunes, R. Kopparapu, C. Cheung, J. Wang, Y. Yu, R. Yin, **B.-H. Wu** et al, CLEO, Charlotte, NC, USA.
- 2026 4. “Bidirectional Nonlinear Optical Tomography: Unbiased Characterization of Off- and On-Chip Coupling Efficiencies,” **B.-H. Wu**, M. J. Mehrabad, D. Englund, Conference on APS Global Physics Summit, Denver, CO, USA.
- 2026 5. “Optimizing the physical front end: microring neuromorphic sensing for weak RF signals,” **B.-H. Wu**, S.-Y. Ma, M. J. Mehrabad, S. K. Vadlamani, H. Choi, D. Englund, SPIE 2026, San Francisco, CA, USA.
- 2025 6. “Quantum Neuromorphic Sensor Network,” **B.-H. Wu**, S.-Y. Ma, S.-Y. Ma, S. K. Vadlamani, H. Choi, D. Englund, Conference on APS Global Physics Summit, Anaheim, CA, USA.
- 2023 7. “Generation of Squeezed Light in Silicon Nitride Photonic Integrated Chips,” S. Liu, A. Hariri, **B.-H. Wu**, Y. Zhang, and Z. Zhang, CLEO, Charlotte, NC, USA.
- 2020 8. “Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications,” **B.-H. Wu**, Z. Zhang and Q. Zhuang, CLEO, San Jose, CA, USA.
- 2020 9. “Quantum-Computing Architecture based on Large-Scale Multi-Dimensional Continuous-Variable Cluster States in a Scalable Photonic Platform,” **B.-H. Wu**, R. N. Alexander, S. Liu, Z. Zhang, Conference on APS March meeting [[Link](#)].

Invited Seminars and Conferences

- 2026, 10. **B.-H. Wu**, “From Continuous-Variable Quantum Information to Intelligent Quantum Photonic Systems,” Stony Brook University, Electrical and Computer Engineering Seminar (**Invited**).
- 2026, 7. **B.-H. Wu**, “Bridging Quantum Photonics and Intelligence: Toward Next-Generation Quantum Technology,” Academia Sinica, Institute of Atomic and Molecular Sciences Seminar (**Invited**).
- 2026, 6. **B.-H. Wu**, “Bridging Quantum Photonics and Intelligence: Toward Next-Generation Quantum Technology,” National Yang Ming Chiao Tung University, Department of Electrophysics Seminar.
- 2026, 5. **B.-H. Wu**, “Breaking On/Off-coupling Loss Degeneracies via Bidirectional Nonlinear Optics,” Conference on Lasers and Electro-Optics Conference, Quantum Measurements & Enabling Technologies session (**Invited**).
- 2026, 3. **B.-H. Wu**, “Bidirectional Nonlinear Optical Tomography: Unbiased Characterization of Off- and On-Chip Coupling Efficiencies,” American Physical Society, Global Physics Summit.
- 2026, 3. **B.-H. Wu**, “Bridging Hardware and Algorithms: Machine Learning for Next-Generation Quantum Sensors,” University of Hawai‘i, Mānoa, Professor and Pizza (**Invited**).
- 2025, 12. **B.-H. Wu**, “Learning Quantum Systems through Physics-Inspired Artificial Intelligence,” University of Hawai‘i, Mānoa, Physics and Astronomy Seminar (**Invited**).
- 2025, 10. **B.-H. Wu**, “Learning Quantum Systems through Physics-Inspired Artificial Intelligence,” University of Hawai‘i, Mānoa, Mathematics Seminar (**Invited**).
- 2025, 4. **B.-H. Wu**, “Physics-Inspired AI for Quantum System Optimization,” University of Hawai‘i, Mānoa, Electrical and Computer Engineering Seminar (**Invited**).
- 2025, 3. **B.-H. Wu**, “Physics-Inspired AI for Quantum System Optimization,” University of Central Florida, Electrical and Computer Engineering Seminar (**Invited**).
- 2025, 3. **B.-H. Wu**, “Advancing Quantum Technologies through Machine-Learning Enhanced Photonic Systems,” Embry-Riddle Aeronautical University, Engineering Physics Seminar (**Invited**).
- 2025, 3. **B.-H. Wu**, “Microring neural sensor in detecting high-speed and low-power radio-frequency signal,”

- American Physical Society, Global Physics Summit.
- 2024, 5. **B.-H. Wu**, “Continuous-variable Photonic Quantum Information,” Cornell University, Applied and Engineering Physics Seminar (**Invited**).
- 2023, 1. **B.-H. Wu**, “Continuous-variable Photonic Quantum Information,” National Tsing Hua University, Physics Seminar.
- 2023, 1. **B.-H. Wu**, “Continuous-variable Photonic Quantum Information,” National Yang Ming Chiao Tung University, Photonics Seminar.
- 2023, 1. **B.-H. Wu**, “Continuous-variable Photonic Quantum Information,” Academia Sinica, Institute of Atomic and Molecular Sciences Seminar.
- 2022, 12. **B.-H. Wu**, “Entanglement-assisted multi-aperture pulse-compression radar for angle resolving detection,” Hon-Hai Research Institute, Quantum Computing Meeting (**Invited**) [[Link](#)].
- 2022, 11. **B.-H. Wu**, “Continuous-variable Photonic Quantum Information,” Massachusetts Institute of Technology, Research Laboratory of Electronics, Optical and Quantum Electronics Seminar (**Invited**) [[Link](#)].
- 2022, 05. **B.-H. Wu**, “Continuous-variable quantum repeaters based on bosonic error-correction and teleportation: architecture and applications,” Conference on Lasers and Electro-Optics Conference, Quantum Communication & Networking session.
- 2020, 03. **B.-H. Wu**, “Quantum-Computing Architecture based on Large-Scale Multi-Dimensional Continuous-Variable Cluster States in a Scalable Photonic Platform,” American Physical Society, March Meeting.